

# MASTER PROFILE REPOSITORY V3 - COMPLETE

James Morseman | Civil Engineering Technology Graduate | Job Search Fact Bank

Prepared June 2026

## Purpose

This document is a comprehensive fact repository for resume tailoring, cover letter generation, interview preparation, and job-search tooling. It is not a resume, not a cover letter, and not a set of scoring instructions. It intentionally preserves details, redundancy, project context, and transferable skills so a tailoring system can select the most relevant verified facts for each application.

Guiding rule: the tool may reframe, select, condense, and reorder facts from this repository, but it should not fabricate experience, inflate credentials, or imply qualifications not supported here.

## 1. Candidate Identity and Positioning

|                                    |                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------|
| <b>Name</b>                        | James Morseman                                                                   |
| <b>Email</b>                       | Jamesmorseman@gmail.com                                                          |
| <b>Phone</b>                       | (631) 559-5622                                                                   |
| <b>Current Region</b>              | New York / Long Island region                                                    |
| <b>Work Authorization</b>          | U.S. citizen                                                                     |
| <b>Driver License</b>              | Valid New York State driver license                                              |
| <b>Current Professional Status</b> | Recent civil engineering technology graduate preparing for FE Civil examination. |

Core positioning: James is a practical, applied civil engineering graduate with a strong structural interest, substantial project leadership experience, hands-on laboratory exposure, and unusually significant operations/personnel-management experience for an entry-level engineering candidate.

Primary differentiators:

- Structural engineering lead and functional project leader for a multidisciplinary capstone project.
- Built and worked with RAM Structural System models for a real educational facility addition project.
- Managed direct reports and operations in a professional environment before entering engineering.
- Developed Excel-based tracking, inventory, accountability, and operational decision-support systems.
- Completed broad civil engineering coursework and labs across structural, geotechnical, construction, water resources, transportation, and materials areas.

## 2. Education

### Farmingdale State College

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| <b>Degree</b>     | Bachelor of Science in Civil Engineering Technology |
| <b>Graduation</b> | May 2026                                            |

|                             |                           |
|-----------------------------|---------------------------|
| <b>Accreditation</b>        | ABET-accredited program   |
| <b>GPA</b>                  | 3.09                      |
| <b>Academic Recognition</b> | Dean's List - Spring 2026 |

Notes: The B.S. Civil Engineering Technology degree should be represented accurately. Do not misstate it as a traditional B.S. Civil Engineering degree. The program is ABET-accredited and should be described as such when relevant.

### Suffolk County Community College

|                          |                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Degree</b>            | Associate of Science in Engineering Science                                                           |
| <b>Completion</b>        | January 2025                                                                                          |
| <b>Program Character</b> | Engineering transfer-oriented program intended for continuation into a four-year engineering program. |

### University at Buffalo

- Previously attended engineering studies prior to Suffolk County Community College and Farmingdale State College.
- Recipient of an academic scholarship while attending University at Buffalo.
- Member of a team that won a freshman engineering design competition for designing the most efficient wind turbine.

## 3. Academic Performance and Coursework

General coursework pool for tailoring. Grades should be stored internally; course grades generally should not be included on a resume unless the tool determines they are strategically useful for a specific application.

### Civil / Structural / Construction Coursework

- Structural Design - AISC steel design, LRFD design methodology, beam and column design.
- Reinforced Concrete Design - concrete beams, columns, footings, development length, shear, flexural, and reinforcement calculations.
- Strength of Materials - stress, strain, shear, bending, deflection, Mohr circle concepts, and mechanics of materials foundations.
- Soils, Foundations, and Earth Structures - geotechnical engineering, soil classification, foundations, bearing behavior, earth structures.
- Construction Materials - concrete, metals, materials testing, strength evaluation, quality-control concepts.
- Field Practices in Civil Engineering - construction planning, safety, permits, equipment selection, scheduling, cost estimation.
- Senior Capstone Design - multidisciplinary engineering design, technical reporting, presentations, and project integration.

### Civil Infrastructure / Water / Transportation Coursework

- Fluid Mechanics / Hydraulics - pipe flow, pressure, head loss, flow measurement, Reynolds number, Bernoulli equation, hydraulic instrumentation.
- Water and Wastewater Systems - stormwater reuse, cistern design, water infrastructure, filtration, pumping, sustainability, maintenance.
- Transportation Engineering - roadway operations, transportation systems, traffic concepts, infrastructure planning.
- Highway Engineering - highway design and transportation infrastructure concepts.

- Surveying - field measurement, site data collection, layout fundamentals, drafting/site interpretation.

### Analytical / Supporting Coursework

- Numerical Methods / Engineering Analysis coursework.
- Operations Research.
- Engineering Economy.
- Technical Writing and engineering report development.
- Calculus, physics, statics, dynamics, and other standard engineering foundations from degree pathway.

## 4. Licensure and Professional Development

- FE Civil Examination scheduled for August 2026.
- Expected next step after FE passage: Engineer-in-Training (EIT) certification.
- Long-term professional goal: Professional Engineer (PE) licensure after qualifying experience.
- Open to graduate-level structural engineering education if it meaningfully advances career goals, especially if employer-supported through tuition reimbursement or related benefits.
- Preference for gaining professional experience before committing to graduate school full time.

## 5. Major Engineering Project: Baldwin High School Instructional Addition

|                         |                                                                             |
|-------------------------|-----------------------------------------------------------------------------|
| <b>Project Type</b>     | Senior civil engineering capstone project                                   |
| <b>Project Name</b>     | Baldwin High School Instructional Addition / Three-Story Classroom Addition |
| <b>Location</b>         | Baldwin, New York                                                           |
| <b>Term</b>             | Spring 2026                                                                 |
| <b>Team</b>             | Horizon Design Group                                                        |
| <b>Role</b>             | Structural Engineering Lead / Project Leader                                |
| <b>Primary Software</b> | RAM Structural System, Revit, Microsoft Office, Excel                       |

Project overview: Multidisciplinary preliminary engineering design for a proposed three-story instructional classroom addition at Baldwin High School. The project addressed structural, geotechnical, architectural, site-civil, drainage, constructability, scheduling, cost, and feasibility considerations for construction adjacent to an active educational facility.

### Leadership and Coordination Responsibilities

- Functioned as Structural Engineering Lead and effectively fulfilled project leader responsibilities throughout significant portions of the project.
- Delegated tasks, coordinated team responsibilities, scheduled meetings, and helped maintain project milestones.
- Managed integration of report sections from multiple disciplines and supported quality control of technical deliverables.
- Coordinated final report development, formatting, editing, appendices, front matter, and submission requirements.
- Authored approximately 60% of final report content; total contribution estimated closer to 75% when edits, appendices, front matter, formatting, and integration are included.
- Presented project findings to students, faculty, and invited industry professionals.
- Capstone report received the highest grade among project groups and was discussed as potential future reference material for subsequent students.

## Structural Engineering Responsibilities

- Built, modified, and analyzed RAM Structural System models with assistance from Kevin Fernandez.
- Performed gravity load development for classroom, hallway, stairwell, and roof loading conditions.
- Performed lateral load development and considered lateral load requirements in the structural system.
- Developed and evaluated structural framing layouts for floor and roof systems.
- Evaluated structural steel as the preferred framing system relative to reinforced concrete alternatives.
- Reviewed and coordinated member geometry, tributary areas, beam line loads, column axial loads, serviceability checks, and preliminary member sizing.
- Extracted structural reactions and member forces from analytical models where needed.
- Coordinated structural decisions with architectural layout, constructability, and foundation requirements.
- Applied ASCE 7-22 loading criteria and AISC steel design methodology in the report context.
- Considered strength, serviceability, deflection behavior, vibration control, member repetition, construction efficiency, and connection simplicity.

## Geotechnical and Foundation Coordination

- Integrated geotechnical information from Carlin-Simpson & Associates reference documents into the preliminary design process.
- Reviewed soil boring, test pit, groundwater, existing fill, and allowable bearing information relevant to foundation design.
- Evaluated shallow foundation feasibility using reinforced concrete spread footings and wall footings bearing on prepared native soil or compacted structural fill.
- Considered uncontrolled fill removal/replacement, medium dense sand bearing conditions, groundwater fluctuation, drainage/dewatering considerations, frost depth, and subgrade preparation.
- Coordinated foundation system assumptions with structural loads and campus constructability constraints.

## Constructability, Schedule, Cost, and Project Delivery

- Evaluated occupied-campus construction constraints, including continued school operations, temporary site access, pedestrian circulation, emergency access, and construction staging limitations.
- Considered phased construction sequencing and site logistics for an active school campus.
- Assisted with cost and feasibility discussions involving structural systems, foundations, grading, drainage, utilities, construction sequencing, and project implementation.
- Maintained integral interaction with nearly all project areas outside direct drainage and earthwork calculations.

## Skills Demonstrated

- Structural analysis and design.
- RAM Structural System modeling.
- Gravity and lateral load development.
- Structural steel framing evaluation.
- Foundation/geotechnical coordination.
- Constructability review.
- Technical report writing.
- Multidisciplinary coordination.
- Project leadership.
- Presentation to technical and non-technical audiences.

## 6. Foundation and Reinforced Concrete Design Project

Associated Project: Baldwin High School Instructional Addition / foundation and footing design work connected to capstone efforts.

- Performed foundation and reinforced concrete design activities tied to a proposed educational facility addition.
- Evaluated shallow spread footing systems and wall footing concepts based on structural loading and geotechnical recommendations.
- Reviewed bearing capacity, settlement, existing fill, groundwater, subgrade preparation, and constructability considerations.
- Coordinated reinforced concrete footing requirements with structural framing, geotechnical conditions, and active-campus construction constraints.
- Considered load transfer mechanisms from structural steel framing into foundations.
- Integrated foundation design recommendations into broader project design documentation.

### Skills Demonstrated

- Foundation design
- Reinforced concrete design
- Footing sizing
- Bearing capacity evaluation
- Soil-structure coordination
- ACI 318 awareness
- Technical documentation

## 7. Structural Steel Design Project - CIV 409

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Course</b>        | CIV 409 Structural Design                      |
| <b>Term</b>          | Summer 2025                                    |
| <b>Project</b>       | Beam EF and Column F project report            |
| <b>Team</b>          | James Morseman, Edwin Muniz, Maritza Estrella  |
| <b>Building Type</b> | Five-story residential building on Long Island |

Project overview: Structural design of Beam EF and Column F as part of the top-floor framing system for a proposed five-story residential building. Design emphasized structural adequacy, material efficiency, economy, and code-compliant steel member selection.

- Performed structural calculations, interpreted design codes, and compiled/wrote the final report.
- Designed Beam EF, a simply supported 20-foot steel beam supporting a 10-foot tributary width.
- Evaluated dead load of 50 psf, live load of 60 psf, and snow load of 15 psf using ASCE 7-16 LRFD load criteria.
- Calculated controlling factored surface load, uniform line load, maximum shear, maximum moment, and allowable deflection.
- Checked flexural strength and serviceability/deflection criteria using AISC Steel Construction Manual, 15th Edition.
- Selected W16x31 steel beam as an efficient section satisfying strength and deflection requirements.
- Designed Column F with a 12-foot height under axial compression and buckling considerations.
- Calculated slenderness ratio, elastic buckling stress, critical buckling stress, nominal compressive strength, and design strength.
- Selected W6x15 steel column as structurally sufficient for axial load and stability requirements.
- Prepared design summary, results, analysis, reflection, conclusion, and calculation appendices.

## Skills Demonstrated

- Steel beam design
- Steel column design
- AISC LRFD methodology
- ASCE 7-16 load development
- Moment and shear calculations
- Deflection/serviceability checks
- Column buckling analysis
- Technical report writing

## 8. Bridge Replacement Construction Management Project - CIV 403W

|                     |                                                                       |
|---------------------|-----------------------------------------------------------------------|
| <b>Course</b>       | CIV 403W Field Practices in Civil Engineering Technology              |
| <b>Project</b>      | Construction Management Project of Bridge Replacement - Spencer       |
| <b>Project Site</b> | NYS Route 34 bridge over Dry Run Creek, Town of Spencer, Tioga County |
| <b>Team</b>         | Andrew Stone, Ava Cutler, James Morseman                              |

Project overview: Construction management planning report for complete replacement of the existing NYS Route 34 bridge over Dry Run Creek, including demolition of the old bridge, construction of a new bridge, roadway approaches, temporary traffic control, equipment, cost estimating, safety, schedule, and lean construction concepts.

- Reviewed project scope involving demolition, replacement bridge construction, roadway approach reconstruction, drainage, interim pavement markings, temporary traffic signals, and traffic control devices.
- Addressed requirement to maintain access to side roads, driveways, and emergency vehicles during construction.
- Developed staff supervision and project management considerations for contractor, transportation engineer, field personnel, subcontractors, safety roles, and compliance oversight.
- Evaluated laws, regulations, and permit requirements relevant to bridge construction and work around waterways.
- Considered environmental controls including streambed diversion, erosion/sediment control, and work-zone impacts.
- Identified heavy equipment needs including excavator, crane, and soil compactor categories.
- Prepared conceptual project cost breakdown and cost-estimation discussion.
- Prepared safety requirements for construction activities, including work-zone hazards, equipment safety, traffic safety, and OSHA-related considerations.
- Developed project schedule summary, construction phases, activity durations, and CPM-style sequencing logic.
- Evaluated lean construction principles applicable to project delivery and field productivity.

## Skills Demonstrated

- Construction management
- Project scheduling
- Cost estimating
- Bridge replacement planning
- Traffic control planning
- Safety planning
- Permitting awareness
- Equipment planning

- Constructability analysis
- Lean construction concepts

## 9. Stormwater Detention / Cistern Design Project - CIV 411

|                |                                                        |
|----------------|--------------------------------------------------------|
| <b>Course</b>  | Water and Wastewater Systems                           |
| <b>Term</b>    | Spring 2026                                            |
| <b>Project</b> | Stormwater Detention / Cistern Design Project          |
| <b>Team</b>    | Avery Fredericks, James Morseman, Richard Phan         |
| <b>Site</b>    | Farmingdale State College - Lupton Hall reference area |

Project overview: Design of a rainwater harvesting and cistern storage system to reuse roof runoff from Lupton Hall for campus landscaping and irrigation, while reducing stormwater discharge into campus drainage infrastructure.

- Identified existing drainage issue: impervious campus coverage prevents effective infiltration and funnels rainwater to storm drains.
- Proposed cistern tanks and pumps to capture, store, filter, and reuse rainwater for landscaping and irrigation.
- Used Lupton Hall as the design reference area and evaluated an 81,000 sf metal roof catchment.
- Applied runoff coefficient of 0.95 and a 2-year, 24-hour storm producing approximately 3.6 inches total rainfall.
- Calculated runoff volume of approximately 23,085 cubic feet, or approximately 173,000 gallons.
- Proposed six cylindrical tanks at roughly 29,000 gallons each, instead of one very large tank.
- Evaluated tank dimensions of approximately 22.2 ft diameter and 10 ft height for each tank.
- Estimated pump capacity at approximately 80-100 gallons per minute and recommended a variable-frequency drive pump to save energy and adjust to demand.
- Included low-water cutoff protection so pump does not operate when tanks are empty.
- Proposed tank locations including Great Lawn, northeast entrance, west entrance, and middle courtyard locations.
- Specified interconnected tanks to equalize water retention and allow pumping from one pump station.
- Included vortex filtration before water enters tanks and secondary filtration before irrigation use.
- Included overflow piping to existing stormwater drains for severe rainfall events.
- Estimated a 173,000 gallon storage system could distribute 1 inch of water over 6 acres in a week; in the project case it could irrigate the Great Lawn for over a week from one major storm.
- Compared detention pond, infiltration trench, green roof, and cistern system alternatives.
- Evaluated limitations including seasonal rainfall variation, irrigation demand, overflow needs, inspection, filter maintenance, winterization, and water quality monitoring.

### Skills Demonstrated

- Stormwater management
- Rainwater harvesting
- Hydrologic calculations
- Cistern design
- Pump evaluation
- Sustainable infrastructure
- System maintenance planning
- Alternative analysis
- Campus infrastructure planning

## 10. Laboratory Experience

### Fluid Mechanics Laboratory - CIV 303

- Pressure Gauge Calibration: calibrated a Bourdon pressure gauge using a dead-weight tester, known piston area, calibrated masses, and pressure comparison across increasing/decreasing pressure trials.
- Responsible in lab for ensuring piston assembly was vertical, properly filled, and suitable for reliable readings; added weights incrementally and helped perform pressure trials.
- Pipe Friction: used H408 Fluid Friction Apparatus on rough pipe test section to measure head loss, flow rates, Reynolds number, velocity, and Darcy-Weisbach friction factor.
- James and Edwin performed majority of experimental setup and operation of apparatus for multiple groups, including adjusting flow control valve, operating collection tank, and preparing pipe section between trials.
- Orifice and Venturi Meters: compared differential pressure flow meters using piezometers, Bernoulli equation, head loss readings, and volumetric flow calculations.
- Evaluated how meter geometry affects energy loss and flow measurement, noting that the orifice meter produced greater head loss while Venturi geometry reduced turbulence and losses.
- Prepared technical reports with tables, figures, plots, calculations, and discussion sections.

### Geotechnical Engineering Laboratory - Soils/Foundations

- Performed specific gravity of soil solids testing using water pycnometer method in accordance with ASTM D854.
- Used oven-dried soil, calibrated pycnometer/flask, deaired water, analytical balance, thermometer, vacuum/hot-plate air removal, and temperature correction concepts.
- Tested light brown clay/sandy soil and evaluated values relative to typical mineral soil ranges.
- Developed experience with soil classification, void ratio/porosity/saturation relevance, and quality requirements for air removal and temperature control.

### Construction Materials Laboratory - CIV 402

- Hardened Concrete Testing: measured concrete cylinder and beam specimens, performed compression testing using neoprene caps and compression testing machine, and calculated compressive strength.
- Concrete compression trials yielded approximately 4,824 psi and 5,014 psi, average roughly 4,919 psi.
- Performed flexural strength testing of concrete beam using four-point bending and calculated modulus of rupture/flexural strength.
- Observed and documented failure modes including Type 6 and Type 3 cylinder breaks and beam crack behavior.
- Charpy Impact Testing: tested brass, aluminum, and steel specimens under hot, room-temperature, and cold conditions.
- Evaluated energy absorption/toughness differences; steel absorbed the greatest impact energy overall, followed by aluminum and brass.
- Operated/supervised testing equipment and participated in group calculations, data organization, and report writing.

### Physics Laboratory Experience

- Buoyant Force / Archimedes Principle lab: determined density of metal object using apparent weight in air and water, displaced volume, buoyant force, and graphical analysis.
- Faraday Law of Induction lab: studied induced EMF, solenoid/wire coil setup, function generator, oscilloscope, DMM, frequency variation, and theoretical relationship between changing magnetic flux and voltage.

## 11. Professional Experience: Urban Air Adventure Park

|                 |                                    |
|-----------------|------------------------------------|
| <b>Employer</b> | Urban Air Adventure Park           |
| <b>Location</b> | Lake Grove, New York               |
| <b>Role</b>     | Event Coordination Department Head |
| <b>Dates</b>    | September 2021 - Present           |

### Staffing and Personnel Management

- Direct reports: approximately 45 employees.
- Typical staffing supervised at a time: approximately 15-25 employees per shift/event period.
- Involved in hiring process and interviews, although not final decision-maker for hiring.
- Participated in performance reviews and employee evaluation processes.
- Provided input regarding staffing decisions and employee performance, although did not have final say on terminations.
- Assisted and played leadership roles in other departments when operational needs required cross-department support.

### Scheduling and Workforce Coordination

- Used Optim8 scheduling software.
- Largest scheduling/staffing coordination involved approximately 55 employees, mostly concentrated on weekends and peak periods while weekdays were lighter.
- Managed scheduling, staffing levels, and operational coverage for event department needs.
- Balanced labor requirements against event volume, business peaks, department coverage, and operational demands.

### Training, Compliance, and Employee Development

- Trained new employees.
- Created or contributed to training procedures.
- Ensured compliance with company procedures and employee requirements.
- Performed or supported testing/evaluation for employees to confirm readiness and understanding.
- Mentored employees and supported team morale.

### Event Coordination and Logistics

- Coordinated birthday parties, school groups, corporate events, and private facility rentals.
- Largest event size exceeded 1,000 guests; approximate maximum around 1,250 guests.
- Managed staffing, logistics, guest flow, event execution, customer expectations, and real-time operational problem solving.
- Worked in a fast-paced environment requiring coordination between front-line employees, managers, guests, and supporting departments.

### Operational Improvement and Excel Systems

- When upper management wanted to eliminate free-entry passes, proposed tracking voucher performance instead of eliminating the program outright.
- Developed Excel-based voucher tracking system to measure how much money each ticket/pass generated after redemption.
- Compared generated value against the potential loss proposition of whether those guests would have returned without free-entry passes or returned as frequently.

- Developed Excel-based inventory tracking system with automated alerts for low supply levels.
- Implemented automated stock reductions at nominal consumption rates when no manual entry was supplied.
- Improved accountability by assigning each employee a specific set of company-provided materials and tracking those materials through the inventory spreadsheet.
- Used Excel as an operational decision-support tool rather than only for basic data entry.

### Transferable Engineering Relevance

- Leadership
- Scheduling
- Workforce management
- Logistics
- Process improvement
- Data tracking
- Excel systems
- Accountability systems
- Operations management
- Training
- Cross-functional coordination
- Real-time problem solving

## 12. Professional Experience: Uncle Giuseppe's Marketplace

|                  |                                           |
|------------------|-------------------------------------------|
| <b>Employer</b>  | Uncle Giuseppe's Marketplace              |
| <b>Locations</b> | Melville, NY; Port Washington, NY         |
| <b>Role</b>      | Assistant Regional Cheese Department Head |
| <b>Dates</b>     | February 2018 - March 2021                |

- Supported multi-location department operations through personnel supervision, scheduling, inventory management, ordering, training, and operational improvements.
- Managed day-to-day department operations in customer-facing, inventory-sensitive environment.
- Served as acting regional department lead during management absences and supported operations across multiple stores.
- Attended supplier/industry showcases and specialized product training activities.
- Traveled to Italy for product training and supplier relationship development.
- Used CAD software to sketch department layout concepts during store redesign efforts before plans were passed to general contractors.
- Received multiple Employee of the Month awards.

## 13. Volunteer Experience

### Dave Wolnitz Foundation

- Volunteer involvement from 2023 to present.
- Role focused on event coordination and logistics for foundation/fundraising events.
- Helped coordinate staffing, activities, raffle/giveaway items, and event setup prior to events.
- During events, often handled more personable public-facing activities such as dealing blackjack table at fundraiser.

- Demonstrated volunteer coordination, logistics, fundraising support, public engagement, event planning, and operations support.

### Guide Dog Foundation

- Volunteered in Summer 2024 to help raise money and support fundraising/outreach activities.

### Farmingdale State College Campus Roadway Cleanup

- Volunteered in Spring 2026 to clean roadways and campus-adjacent areas around Farmingdale State College.

## 14. Organizations, Activities, and Memberships

- ASCE Student Chapter: attended many Farmingdale ASCE student chapter meetings; formal dues-paying membership not confirmed.
- Suffolk County Community College Club Lacrosse: member of club lacrosse team.
- Rocketry Club: member/participant; fabricated rockets from scratch using cylindrical steel and sheet steel for fins/components.
- Astronomy Club: member/participant.
- Pi Kappa Phi fraternity: member of social fraternity.

## 15. Awards and Recognition

- Dean's List - Farmingdale State College, Spring 2026.
- Highest-graded capstone report among capstone groups; faculty suggested it could be used as future reference material.
- Multiple Employee of the Month awards at Urban Air / prior professional experience context.
- University at Buffalo wind turbine efficiency design competition winner as part of engineering design group in 2015.
- University at Buffalo academic scholarship recipient.
- AP Scholar with Distinction from high school; stored for completeness, generally not prioritized for engineering resumes.

## 16. Personal Engineering and Technical Projects

### Custom Standing Desk Design and Fabrication

- Built custom desk from butcher block materials.
- Designed the desk around tight room space constraints using CAD/Autodesk Inventor.
- Stained components and performed hands-on fabrication/assembly.
- Designed and fabricated custom feet/caps/components using Inventor and a friend's 3D printer.
- Upgraded the desk over time to include standing-desk functionality.
- Demonstrates CAD, design constraints, fabrication, iterative improvement, and practical problem solving.

### Rocketry Fabrication Projects

- Fabricated several rockets from scratch through Rocketry Club.
- Worked with cylindrical steel and sheet steel for fins and other components.
- Demonstrates hands-on fabrication, aerospace/rocketry interest, technical curiosity, and teamwork.

## Custom Gaming PC Build

- Selected compatible components and assembled a custom gaming PC.
- Performed system configuration, troubleshooting, and testing.
- Demonstrates hardware integration, component selection, technical troubleshooting, and technology interest.

## Mechanical LEGO Engine Models

- Built LEGO engine/mechanical assemblies demonstrating mechanical visualization, assembly, and interest in moving systems.

# 17. Software and Technical Skills

## Verified Software Proficiency

|                              |                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AutoCAD</b>               | Intermediate. Used for drafting/design-related coursework and technical layout work.                                                                                 |
| <b>Revit</b>                 | Beginner / basic working knowledge. Used or exposed through academic/capstone design coordination.                                                                   |
| <b>RAM Structural System</b> | Intermediate. Used for capstone structural modeling, gravity/lateral load evaluation, member force/reaction extraction, and framing assessment.                      |
| <b>Autodesk Inventor</b>     | Intermediate. Used for personal fabrication/design projects and CAD component modeling.                                                                              |
| <b>Microsoft Project</b>     | Basic working knowledge. Used for project planning/scheduling context.                                                                                               |
| <b>Microsoft Excel</b>       | Advanced. Used for engineering calculations and professional operations systems including voucher tracking, inventory, automated alerts, and accountability systems. |
| <b>MATLAB</b>                | Basic working knowledge.                                                                                                                                             |
| <b>AI Productivity Tools</b> | Intermediate. Used for research support, technical writing, brainstorming, workflow optimization, document development, and project support.                         |

## Software Learning Profile

- Willing and able to learn discipline-specific software where required or beneficial.
- Adjacent software considered learnable: Civil 3D, Bluebeam, ETABS, SAP2000, RISA, STAAD, MicroStation, HEC-RAS, GIS, HydroCAD, Primavera P6, Procore, Navisworks.
- Do not claim direct experience with software not verified above, but it is appropriate to state demonstrated ability to learn new engineering software.

# 18. Career Interests and Preferences

- Structural Engineering interest weight: approximately 40%. Interest in building structures, structural analysis, steel design, concrete/foundations, constructability, and structural systems.
- Construction / Project Engineering interest weight: approximately 35%. Interest in project delivery, scheduling, logistics, field coordination, constructability, and management of technical work.
- Site Civil / Land Development interest weight: approximately 25%. Open to infrastructure, transportation, site-civil coordination, stormwater, utilities, grading, and municipal work.

- Primary goal for first role is to gain meaningful engineering experience under experienced professionals, build technical competency, and progress toward EIT/PE path.
- Open to structural graduate education later, especially if supported by employer tuition assistance.

## 19. Geographic, Work Arrangement, and Compensation Philosophy

- Open to relocation throughout the United States if opportunity quality justifies relocation.
- Not vehemently opposed to relocating anywhere if the opportunity is strong enough.
- Geographic scoring is expected to be informed by separate city/metro analysis document.
- Coastal regions, major bodies of water, mountainous regions, NYC metro, LA/Southern California, and high-opportunity civil engineering markets may receive preference, but should not be treated as hard requirements.
- Commutes up to approximately one hour are acceptable but should be treated as a scoring drawback relative to shorter commutes.
- Hybrid work preferred but not required.
- Overtime opportunities viewed positively but not required.
- Salary floor is better represented by projected savings/financial feasibility rather than raw salary alone.
- Cannabis-related location preferences are candidate-side only and should never be employer-facing or implied in employer documents.

## 20. Languages and Additional Attributes

- Native English speaker.
- No formal second-language proficiency claimed.
- Can understand Spanish fairly well, especially reading/listening comprehension; should not be overstated as professional fluency.
- Valid New York State driver license.
- No internships completed; candidate prioritized accelerated graduation through summer and intersession coursework as an older/nontraditional student seeking to complete degree efficiently. Use this explanation only when strategically appropriate; do not volunteer it unnecessarily on resumes.

## 21. Keywords and Evidence Pool

This section stores searchable keyword phrases tied to verified experience. It is not meant to be pasted wholesale into resumes.

- Structural: structural analysis, steel design, reinforced concrete design, RAM Structural System, gravity loads, lateral loads, AISC, ASCE 7, framing layout, beam design, column design, footing design, spread footings, wall footings, bearing capacity, deflection, serviceability, member sizing.
- Construction: construction planning, project scheduling, CPM, cost estimating, safety planning, traffic control, equipment selection, permitting, bridge replacement, constructability, phasing, occupied campus, logistics, field coordination.
- Water/Site Civil: stormwater management, rainwater harvesting, cistern design, runoff calculations, hydrology, irrigation reuse, pumping system, filtration, overflow, sustainable infrastructure, campus integration.
- Laboratory: pressure gauge calibration, dead-weight testing, pipe friction, head loss, Reynolds number, Venturi meter, orifice meter, Bernoulli equation, soil specific gravity, ASTM D854, concrete compression, flexural strength, Charpy impact testing, failure analysis.

- Leadership/Operations: direct reports, scheduling, training, interviews, performance reviews, Optim8, event logistics, process improvement, Excel tracking, inventory management, automated alerts, accountability systems, high-volume events, 1,250 guests.

END OF MASTER PROFILE REPOSITORY V3 COMPLETE